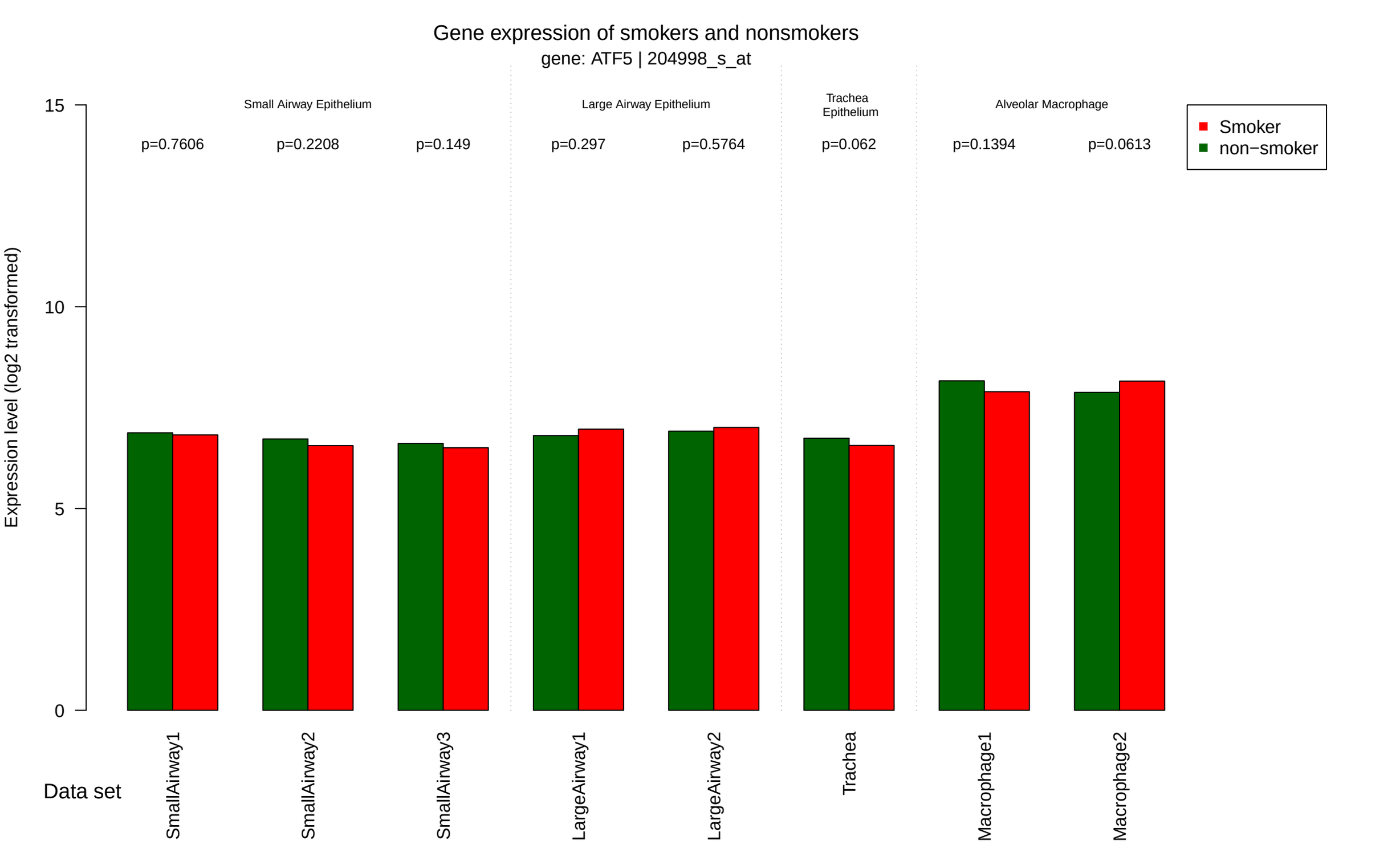
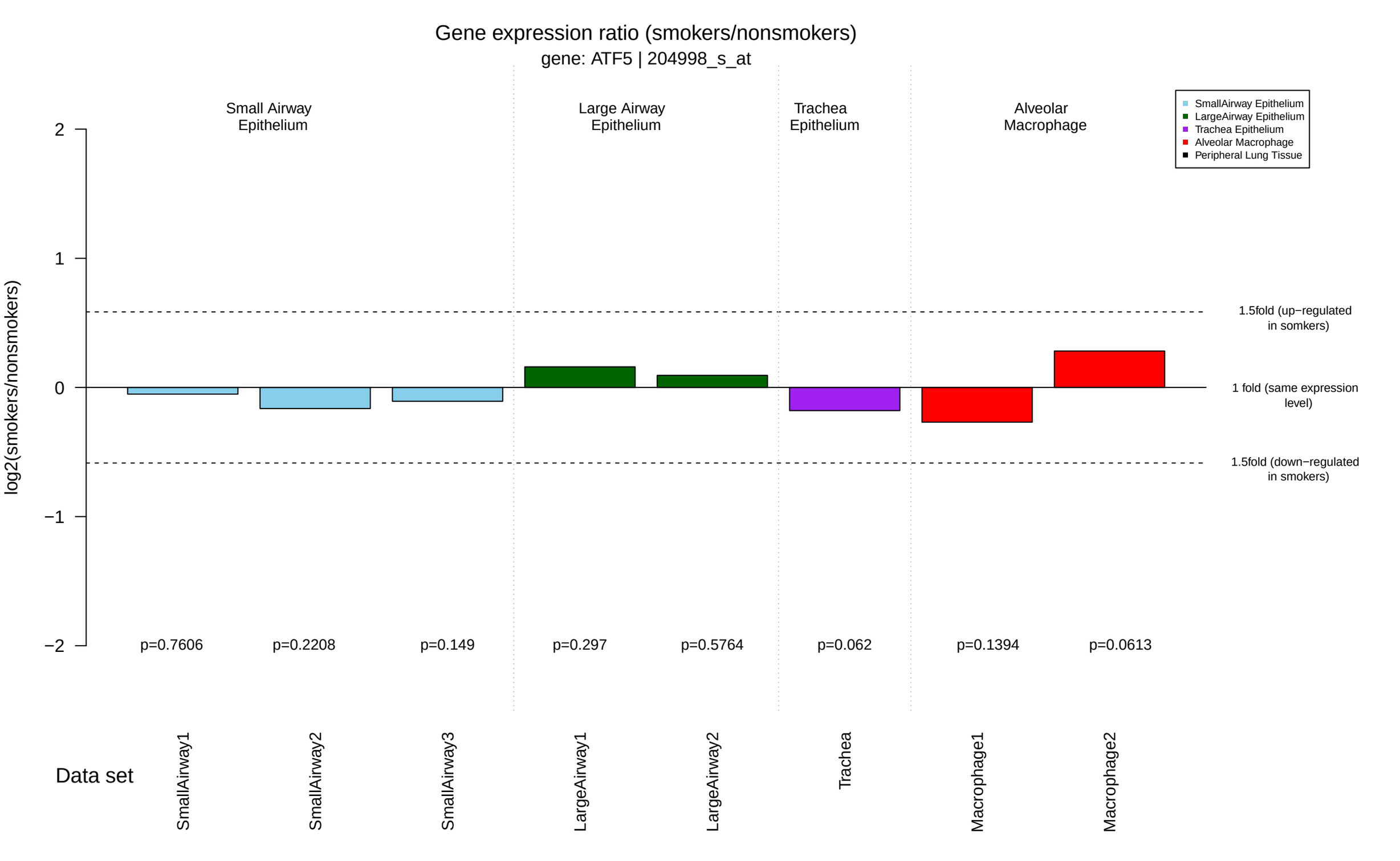
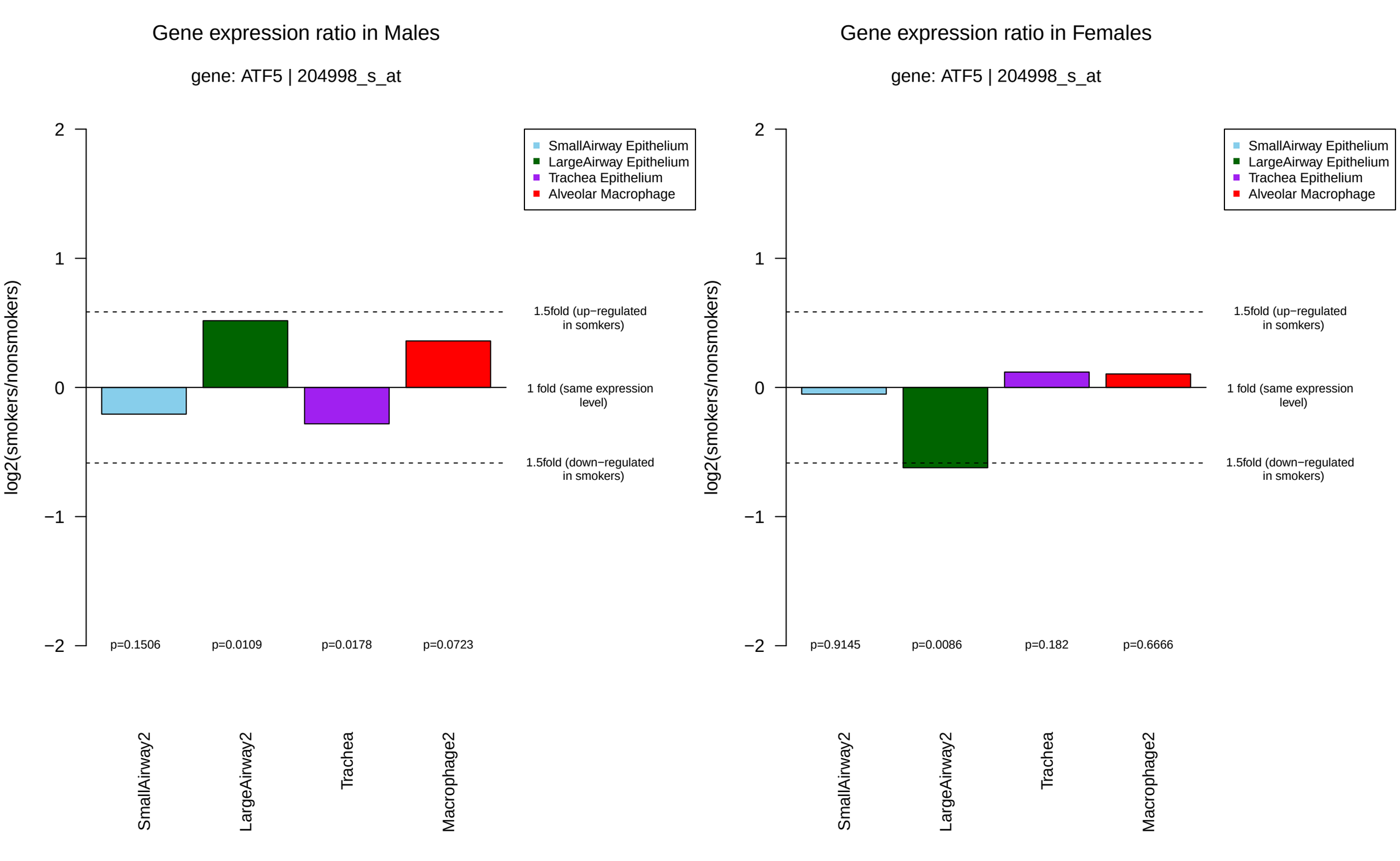
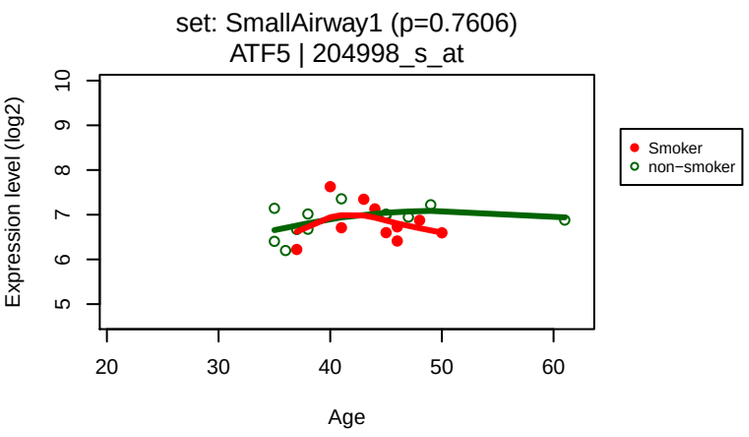
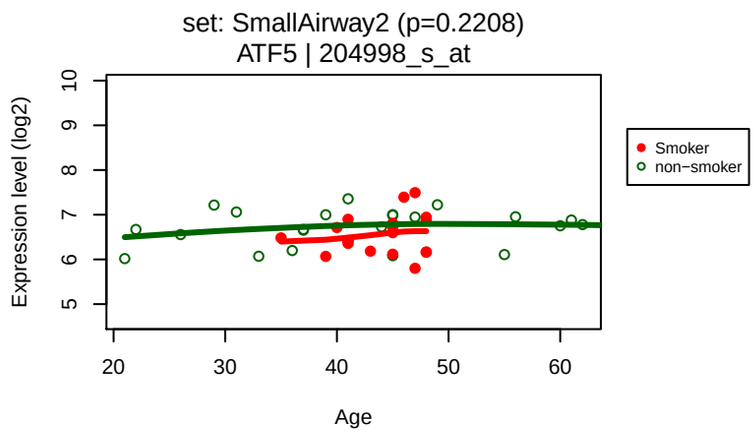
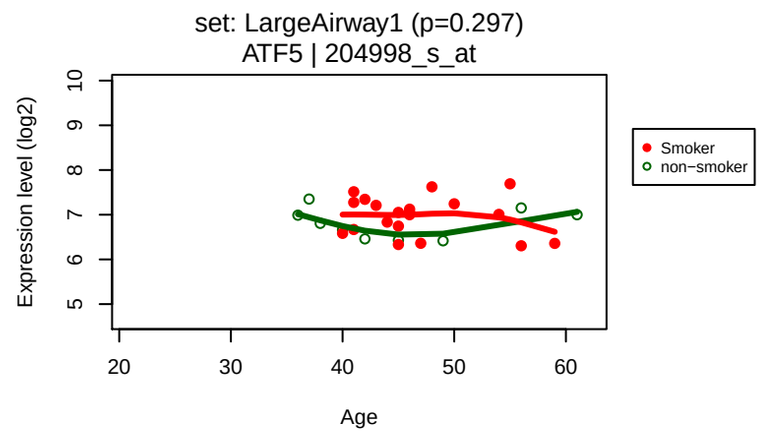

Gene expression of smokers and nonsmokers
gene: ATF5 | 204998_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7606
p=0.2208
p=0.149
p=0.297
p=0.5764
p=0.062
p=0.1394
p=0.0613
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

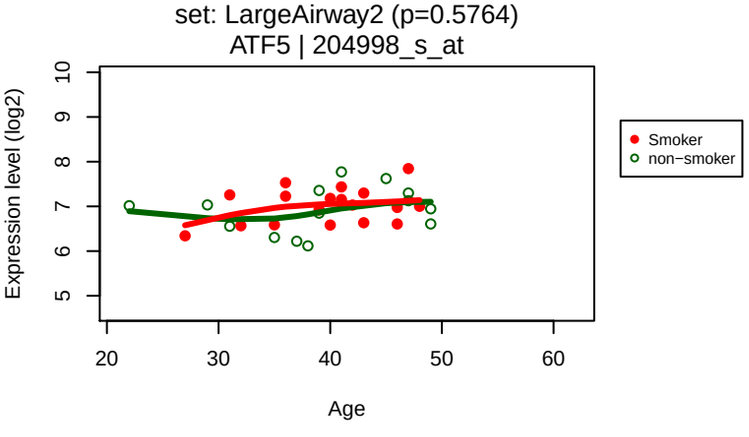

Gene expression ratio (smokers/nonsmokers)
gene: ATF5 | 204998_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7606
p=0.2208
p=0.149
p=0.297
p=0.5764
p=0.062
p=0.1394
p=0.0613
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

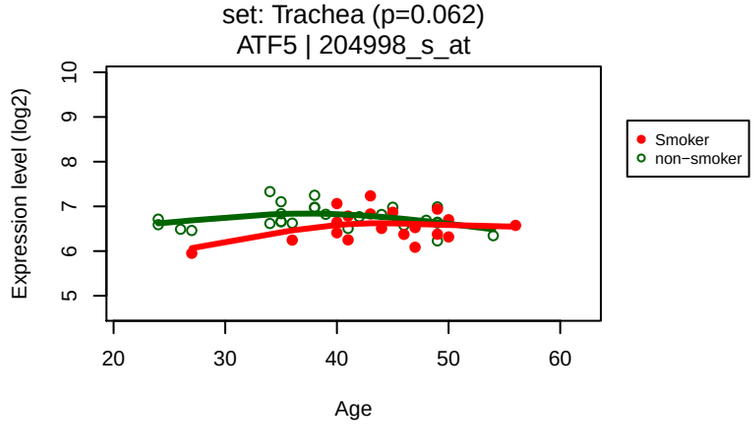

Gene expression ratio in Males
gene: ATF5 | 204998_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1506
p=0.0109
p=0.0178
p=0.0723
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ATF5 | 204998_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9145
p=0.0086
p=0.182
p=0.6666
SmallAirway2
LargeAirway2
Trachea
Macrophage2

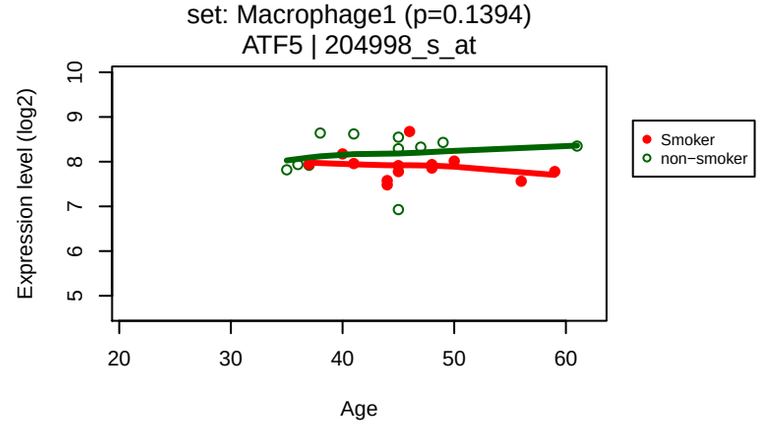

set: SmallAirway1 (p=0.7606)
ATF5 | 204998_s_at
Expression level (log2)
Age
Smoker
non−smoker

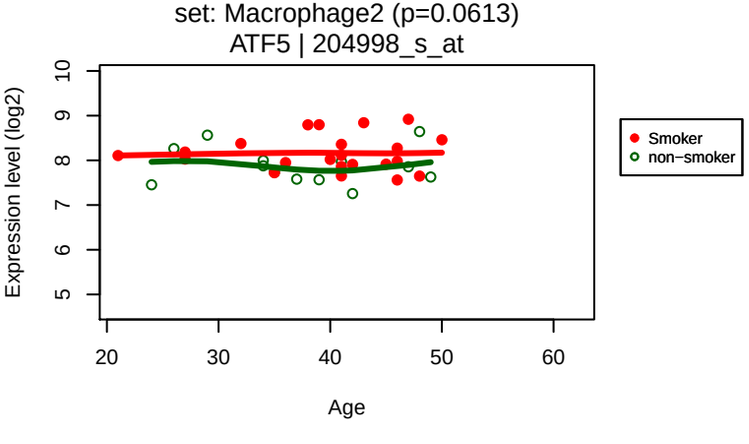

set: SmallAirway2 (p=0.2208)
ATF5 | 204998_s_at
Expression level (log2)
Age
Smoker
non−smoker

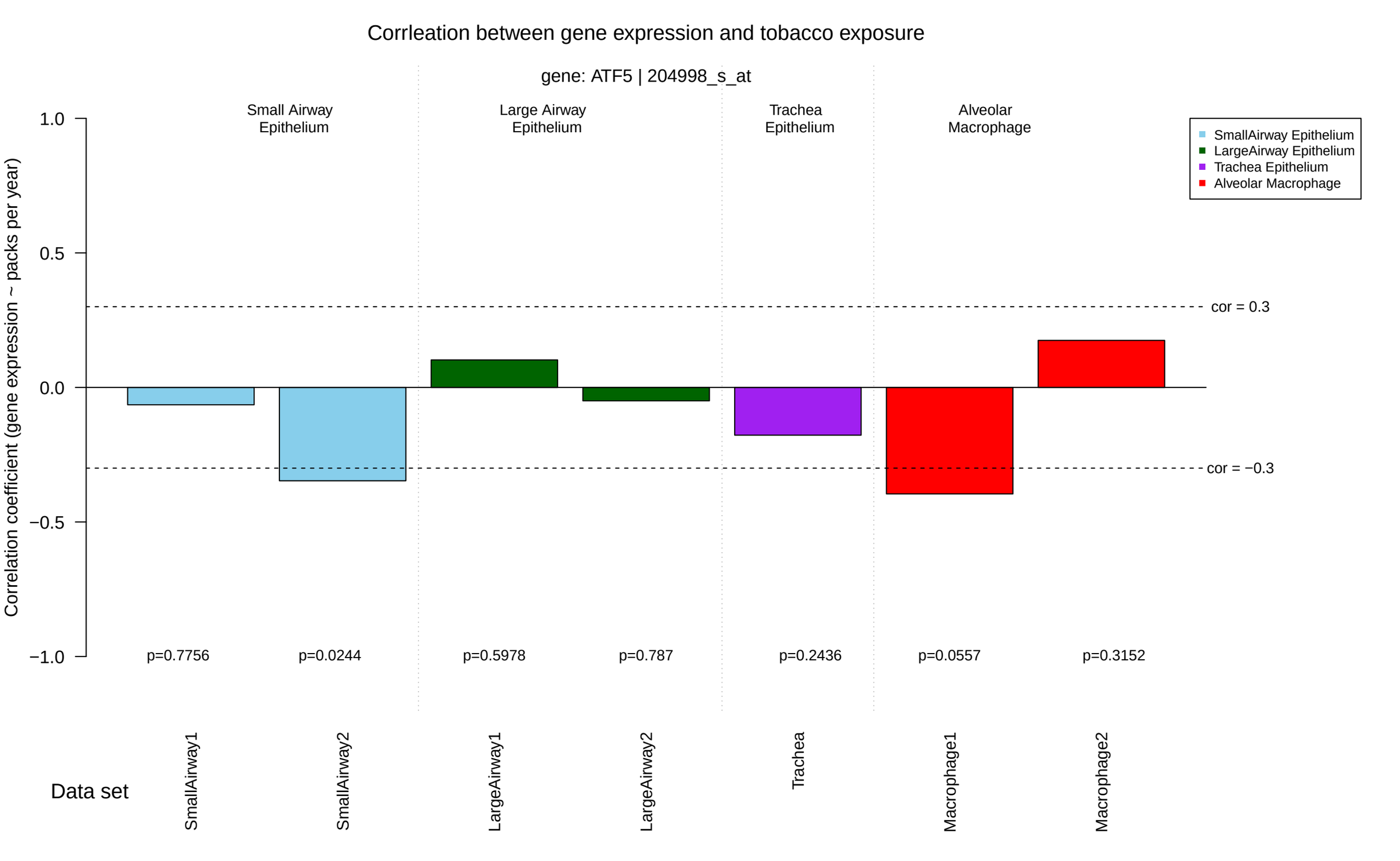

set: LargeAirway1 (p=0.297)
ATF5 | 204998_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5764)
ATF5 | 204998_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.062)
ATF5 | 204998_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1394)
ATF5 | 204998_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0613)
ATF5 | 204998_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ATF5 | 204998_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7756
p=0.0244
p=0.5978
p=0.787
p=0.2436
p=0.0557
p=0.3152
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2